

Chemistry, 2017 EC, entrance exam, version 3

Grade 12 · Chemistry · 2017 E.C.

Time allowed: 120 minutes

Questions: 76

Total marks: 76

Name _____ Grade _____ ID No. _____ Date _____

1. According to the Brønsted-Lowry theory, a base is defined as: (1)
A. A proton donor
B. An electron-pair donor
C. A proton acceptor
D. An electron-pair acceptor
2. Which of the following solid is NOT amorphous solid? (1)
A. Glass
B. Plastics
C. Rubber
D. Sodium chloride
3. In an ionic bond, electrons are: (1)
A. Shared equally between atoms.
B. Transferred from one atom to another.
C. Shared unequally between atoms.
D. Completely absent.
4. Which of the following is true about the liquid state of matter? (1)
A. Liquids have a definite volume but take the shape of their container
B. Liquids have a definite shape and volume
C. Liquids have an indefinite volume but a definite shape
D. Liquids have neither a definite shape nor volume
5. What is true for the following given reaction? $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{Cl}^- \rightarrow 2\text{Cr}^{3+} + 3\text{Cl}_2 + 7\text{H}_2\text{O}$ (1)
A. Cr^{3+} is oxidized
B. Cl^- is reduced to Cl_2
C. H^+ is reduced to H_2
D. Cl^- is oxidized to Cl_2
6. What is the conjugate base of H_2CO_3 ? (1)
A. HCO_3^-
B. CO_3^{2-}
C. OH^-
D. H_3O^+
7. Which gas law relates the pressure and volume of a gas at constant temperature? (1)
A. Boyle's Law
B. Charles's Law
C. Gay-Lussac's Law
D. Avogadro's Law
8. Which of the following best describes metallic bonding? (1)
A. Electrons are shared equally between atoms
B. Electrons are transferred from one atom to another
C. Electrons are delocalized and free to move throughout the metal
D. Electrons are localized around each atom
9. Given a 0.1 M solution of a weak acid ($K_a = 1.8 \times 10^{-5}$), what is the approximate pH of the solution? (1)
A. 2.87
B. 3.45
C. 4.72
D. 5.60

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10. What happens to the kinetic energy of particles as a substance moves from a solid to a liquid state? (1)
- A. The kinetic energy decreases B. The kinetic energy increases
C. The kinetic energy remains the same D. The particles stop moving
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11. What type of intermolecular force is responsible for the high boiling point of water? (1)
- A. London dispersion forces B. Hydrogen bonds
C. Ionic bonds D. Dipole-dipole interactions
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12. Why does a buffer solution resist changes in pH? Because (1)
- A. it contains only a strong acid
B. it contains a weak acid and its conjugate base
C. it prevents ionization of water
D. it absorbs all added ions
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13. What is the wavelength of the yellow sodium emission, which has a frequency of $5.09 \times 10^{14} \text{ s}^{-1}$? (1)
- A. 890 nm B. 400 nm
C. 589 nm D. 670 nm
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14. In a covalent bond, how do atoms achieve a stable electron configuration? (1)
- A. By transferring electrons from one atom to another.
B. By sharing electrons.
C. By losing all their electrons.
D. By gaining electrons from other elements.
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15. Which of the following factors affects the ionization of weak acids? (1)
- A. Temperature B. Pressure
C. Volume D. Mass
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16. The blue color in fireworks is often achieved by heating copper(I) chloride to about 1200°C . Then the compound emits blue light having a wavelength of 450 nm. What is the increment of energy (the quantum) that is emitted at $4.50 \times 10^2 \text{ nm}$ by CuCl ? (1)
- A. $9.93 \times 10^{-19} \text{ J}$ B. $4.42 \times 10^{-19} \text{ J}$
C. $12.1 \times 10^{-2} \text{ J}$ D. $6.1 \times 10^{-5} \text{ J}$
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17. In a molecule of nitrogen (N_2), how many electrons are shared between the two nitrogen atoms? (1)
- A. 2 electrons B. 4 electrons
C. 6 electrons D. 10 electrons
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18. If the reaction, $\text{Cu} + \text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}_2 + \text{H}_2\text{O}$ is balanced. What is the coefficient of HNO_3 ? (1)
- A. 8 B. 3
C. 2 D. 4
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19. A sample of gas occupies 10.0 L at 2.0 atm pressure. What will be its volume if the pressure is reduced to 1.0 atm at constant temperature? (1)
- A. 5.0 L B. 20.0 L
C. 15.0 L D. 10.0 L

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- 20.** A salt derived from a weak acid and a weak base will produce a solution whose pH depends on: (1)
- A. The strength of the weak acid only
 - B. The strength of the weak base only
 - C. The relative strengths of the weak acid and weak base
 - D. The concentration of the salt
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- 21.** At STP, 2 moles of an ideal gas will occupy: (1)
- A. 11.2 L
 - B. 22.4 L
 - C. 44.8 L
 - D. 33.6 L
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- 22.** A student electrolyzes an aqueous copper sulfate solution using platinum electrodes. What will they observe at the anode? (1)
- A. Copper deposition
 - B. Hydrogen gas evolution
 - C. Oxygen gas evolution
 - D. Chlorine gas evolution
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- 23.** If a titration curve shows a sharp pH rise around pH 9, what type of titration is it? (1)
- A. Strong acid vs. strong base
 - B. Strong acid vs. weak base
 - C. Weak acid vs. strong base
 - D. Weak acid vs. weak base
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- 24.** Which of the following is the correct order of elements in the Modern Periodic Table? (1)
- A. Elements are arranged by increasing atomic number.
 - B. Elements are arranged by increasing atomic mass.
 - C. Elements are arranged by increasing chemical properties.
 - D. Elements are arranged by their colors.
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- 25.** Which law states that mass is neither created nor destroyed in a chemical reaction? (1)
- A. Law of Definite Proportions
 - B. Law of Conservation of Mass
 - C. Law of Multiple Proportions
 - D. Dalton's Law
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- 26.** Which of the following is a key characteristic of chemical industries? (1)
- A. They avoid the use of chemical reactions
 - B. They produce only organic compounds
 - C. They consume large quantities of energy
 - D. They do not test product quality
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- 27.** What is a major environmental concern related to cement manufacturing? (1)
- A. Excessive water use
 - B. Carbon dioxide emissions
 - C. Excessive gypsum use
 - D. Noise pollution
-
- 28.** According to Dalton's atomic theory, atoms of the same element are: (1)
- A. Different in size and mass
 - B. Always in motion
 - C. Identical in size and mass
 - D. Made up of smaller particles
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- 29.** What is the main raw material obtained from the atmosphere for industrial use? (1)
- A. Nitrogen gas
 - B. Sodium chloride
 - C. Phosphates
 - D. Crude petroleum

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- 30.** One of the advantages of periodic classification is that it helps in understanding: (1)
- A. The periodicity of chemical properties among elements
 - B. The exact uses of each element in everyday life
 - C. The number of atoms in a compound.
 - D. The physical appearance of each element.
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- 31.** Which subatomic particle has a mass closest to that of a proton? (1)
- A. Electron
 - B. Photon
 - C. Neutron
 - D. Positron
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- 32.** What role does industrial chemistry play in society? (1)
- A. Reduces technological development
 - B. Limits chemical product manufacturing
 - C. Prevents resource utilization
 - D. Supports economic growth and industrial development
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- 33.** Which of the following elements is correctly matched with its block in the periodic table? (1)
- A. Na, Mg, and Cl - s-block
 - B. Fe, Mn and P - d-block
 - C. P, S and Cl - p-block
 - D. Cr, Ni and Al - p-block
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- 34.** What does the Aufbau principle state about the filling of electron orbitals? (1)
- A. Electrons will fill the orbitals in order of increasing energy.
 - B. Electrons will fill the orbitals in order of increasing atomic number.
 - C. Electrons will fill the orbitals from higher to lower energy
 - D. Electrons will fill the orbitals randomly.
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- 35.** Which natural resource is most crucial for energy production in developing countries? (1)
- A. Biomass
 - B. Crude oil
 - C. Hydropower
 - D. Solar energy
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- 36.** Which industry significantly contributes to Ethiopia's agricultural economy? (1)
- A. Sugar industry
 - B. Cement industry
 - C. Glass industry
 - D. Paper industry
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- 37.** Which of the following best describes the Rutherford gold foil experiment? (1)
- A. Discovered the electron
 - B. Showed that atoms are mostly empty space with a dense nucleus
 - C. Proposed the planetary model of the atom
 - D. Demonstrated the law of multiple proportions
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- 38.** What is the primary industrial process for manufacturing ammonia? (1)
- A. Ostwald process
 - B. Solvay process
 - C. Haber-Bosch process
 - D. Contact process

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- 39.** Why does the electron configuration of chromium (Cr, atomic number 24) deviate from the expected $[\text{Ar}] 4s^2 3d^4$? (1)
- A. Chromium prefers a half-filled 3d subshell for stability.
 - B. Chromium has fewer protons than expected.
 - C. Chromium's 4s orbital is filled before the 3d.
 - D. Chromium's 4p orbital is partially filled.
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- 40.** Which renewable resource can directly be used for generating electricity? (1)
- A. Solar energy
 - B. Coal
 - C. Crude petroleum
 - D. Magnesium
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- 41.** Which of the following is the correct value of the angular momentum quantum number (l) for a d-orbital? (1)
- A. 2
 - B. 1
 - C. 0
 - D. 3
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- 42.** Which resource is classified as non-renewable? (1)
- A. Fossil fuels
 - B. Vegetation
 - C. Solar energy
 - D. Water
-
- 43.** How many unpaired electrons are present in the electron configuration of carbon ($1s^2 2s^2 2p^2$)? (1)
- A. 1
 - B. 3
 - C. 4
 - D. 2
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- 44.** Which raw material is primarily used in glass manufacturing? (1)
- A. Limestone
 - B. Silica
 - C. Gypsum
 - D. Alumina
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- 45.** Which of the following forms of electromagnetic radiation has the shortest wavelength? (1)
- A. Radio waves
 - B. Infrared waves
 - C. Ultraviolet waves
 - D. Gamma rays
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- 46.** Which local Ethiopian alcoholic beverage is commonly made using traditional methods? (1)
- A. Wine
 - B. Beer
 - C. Vodka
 - D. Araki
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- 47.** What is used in the Contact Process for sulfuric acid production? (1)
- A. Iron oxide
 - B. Vanadium (V) oxide
 - C. Platinum
 - D. Palladium
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- 48.** Which method is commonly used to preserve food in Ethiopia? (1)
- A. Vacuum packing
 - B. Chemical preservation
 - C. Sun drying
 - D. Pasteurization
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- 49.** What is the main purpose of tanning in leather production? (1)
- A. To make leather waterproof
 - B. To remove hair from hides
 - C. To improve the color of leather
 - D. To convert raw hides into durable leather

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- 59.** A student is asked to balance the redox reaction using the half-reaction method. What is the first step? (1)
- A. Balance oxygen atoms
 - B. Identify oxidation and reduction half-reactions
 - C. Add water molecules
 - D. Balance hydrogen atoms
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- 60.** Who discovered the nucleus of an atom? (1)
- A. Niels Bohr
 - B. J.J. Thomson
 - C. James Chadwick
 - D. Ernest Rutherford
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- 61.** What determines the ions preferentially discharged during electrolysis? (1)
- A. Temperature
 - B. Reactivity series and ion concentration
 - C. Voltage applied
 - D. Electrode size
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- 62.** According to Bohr's atomic model, where are electrons located? (1)
- A. Embedded in a positive sphere
 - B. Randomly distributed inside the nucleus
 - C. Fixed orbits around the nucleus
 - D. In stationary energy clouds
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- 63.** What happens if the salt bridge in a voltaic cell is removed? (1)
- A. The voltage increases
 - B. The reaction continues unaffected
 - C. The flow of ions stops and the cell stops working
 - D. The electrodes dissolve
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- 64.** Design an experiment to determine the mass of silver deposited when a current of 2 A is passed through silver nitrate for 30 minutes. What formula would you use? (1)
- A. $m = V i t$
 - B. $m = M I t / n F$
 - C. $m = z Q$
 - D. $m = n F$
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- 65.** Why do ionic compounds have high melting points? (1)
- A. They have weak intermolecular forces
 - B. They consist of covalent bonds
 - C. They have no molecular structure
 - D. The strong electrostatic attraction between ions requires a large amount of energy to break
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- 66.** Who proposed the first atomic theory based on experimental evidence? (1)
- A. J.J. Thomson
 - B. Niels Bohr
 - C. Ernest Rutherford
 - D. John Dalton
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- 67.** Which of the following is NOT a common industrial application of electrolysis? (1)
- A. Electroplating
 - B. Electro-refining
 - C. Metal extraction
 - D. Nuclear fusion
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- 68.** Which of the following molecules would likely exhibit sp^3 hybridization in its central atom? (1)
- A. CO_2
 - B. BF_3
 - C. H_2O
 - D. PCl_5

69. Which model of the atom is referred to as the 'plum pudding model'? (1)

- A. Thomson's Model** **B. Dalton's Model**
C. Rutherford's Model **D. Bohr's Model**

70. Which bonding theory explains the formation of covalent bonds through the overlap of atomic orbitals? (1)

- A.** Valence bond theory
B. Molecular orbital theory
C. Lewis dot theory
D. VSEPR theory

71. Which of the following molecules has a linear molecular geometry? (1)

- A.** Water (H_2O)
- B.** Ammonia (NH_3)
- C.** Methane (CH_4)
- D.** Carbon dioxide (CO_2)

72. Create a cell diagram for a voltaic cell using Zn and Cu electrodes in their respective sulfate solutions. How will it be represented? (1)

- A. $\text{Zn(s)} / \text{Cu}^{2+}(\text{aq}) // \text{Cu}^{2+}(\text{aq}) / \text{Cu(s)}$
- B. $\text{Cu(s)} / \text{Cu}^{2+}(\text{aq}) // \text{Zn}^{2+}(\text{aq}) / \text{Zn(s)}$
- C. $\text{Zn}^{2+}(\text{aq}) / \text{Zn(s)} // \text{Cu(s)} / \text{Cu}^{2+}(\text{aq})$
- D. $\text{Zn(s)} / \text{Zn}^{2+}(\text{aq}) // \text{Cu}^{2+}(\text{aq}) / \text{Cu(s)}$

73. Which type of covalent bond is formed when electrons are shared equally between two atoms? (1)

- A.** Polar covalent bond **B.** Ionic bond
C. Non-polar covalent bond **D.** Coordinate bond

74. What is the hybridization of the central atom in a molecule of carbon dioxide (CO_2)? (1)

- A. sp^2
B. sp^3d
C. sp
D. sp^3

75. A current of 5 A flows for 2 hours during electrolysis. What is the total charge passed? (1)

- A. 180 C B. 3600 C
C. 1,800 C D. 36,000 C

76. What is the bond angle in a molecule with a tetrahedral geometry, such as methane (CH_4)? (1)

- A. 90°
B. 120°
C. 180°
D. 109.5°

— END OF EXAMINATION —